

Esolution

Place student sticker here

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
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Statistics for Business Administration

Exam: MA9712 / Retake
Examiner: Prof. Mathias Drton

Date: Friday 6th October, 2023
Time: 13:30 – 15:00

Working instructions

- This exam consists of **16 pages** with a total of **8 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 44 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **scientific pocket calculator**
 - the **cheat sheet**, which is contained in this exam on page **15**
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Additional space for solution to **any** of the problems is given on pages 14 and 16.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____

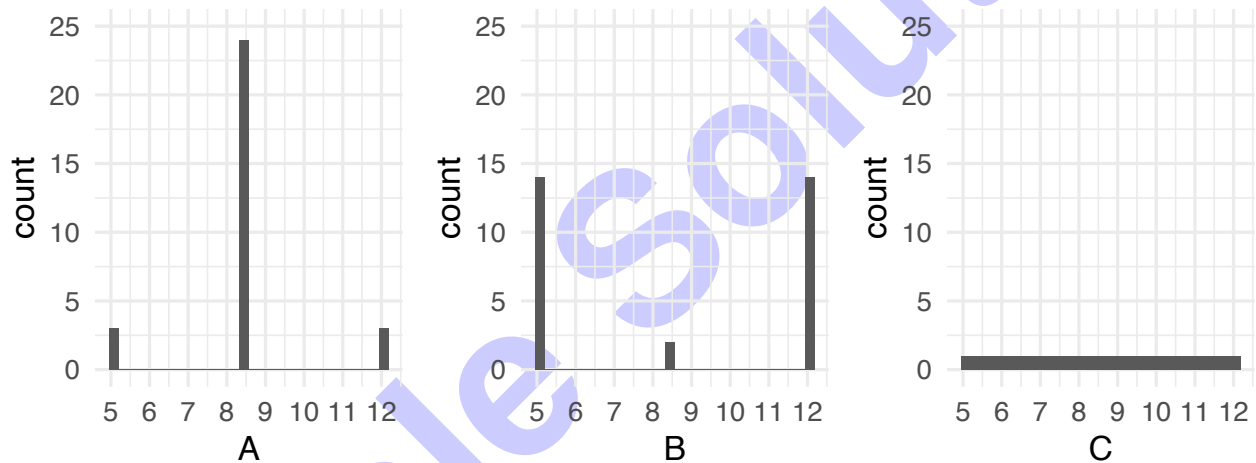
Problem 1 (5 credits)

Give a short answer to each of the following **unrelated** sub-problems.

- 0 ☐
1 ☐
- a) In a study of three nationally representative large-scale data sets from Ireland, the United States, and the United Kingdom ($n = 17247$), teenagers between the ages of 12 to 15 were asked to keep a diary of their screen time and answer questions about how they felt or acted. The answers to these questions were then used to compute a psychological well-being score. Additional data were collected, and included in the analysis, such as each child's sex and age, and on the mother's education, ethnicity, psychological distress, and employment. The study concluded that there is clear evidence that screen time correlates negatively with adolescent well-being.

Can the results of the study be used to establish causal relationships? Justify your answer.

- 0 ☐
1 ☐
- b) Below you see a histogram for each one of the three data sets: A, B and C.



Order the empirical standard deviations s_A , s_B and s_C from **smallest to largest**. You **do not need** to justify your answer.

- 0 ☐
1 ☐
- c) A professor grades a test for 140 students and computes the empirical median of all 140 test results, which is 41. She then realizes that the five students with the highest scores did exceptionally well, and she decides to award these five students a bonus of 10 more points. Determine the empirical median of the adjusted test scores. Justify your answer.

d) A 2015 poll conducted by the Pew Research Center found that 15% of American adults have used online dating sites or mobile dating apps. What is the probability that in a random sample of 20 adults, at least one has used online dating sites or mobile dating apps?

☐ 0
☐
☐ 1

e) Consider the following card game with a well-shuffled deck of 52 cards (consisting of 13 ranks in each of the four suits: clubs, diamonds, hearts, and spades). If you draw a red card, you win nothing. If you get a spade, you win \$5. For any club, you win \$10 plus an extra \$30 for the ace of clubs. Find the expected winnings for a single game.

☐ 0
☐
☐ 1

Problem 2 (3 credits)

About 24% of human twins are identical, and the rest are fraternal. Identical twins are necessarily the same sex – half are males and the other half are females. 28% of fraternal twins are both male, 27% both female, and 45% are mixes: one male, one female.

a) You have just become a parent of twins and are told they are both girls. Given this information, what is the probability that they are identical?

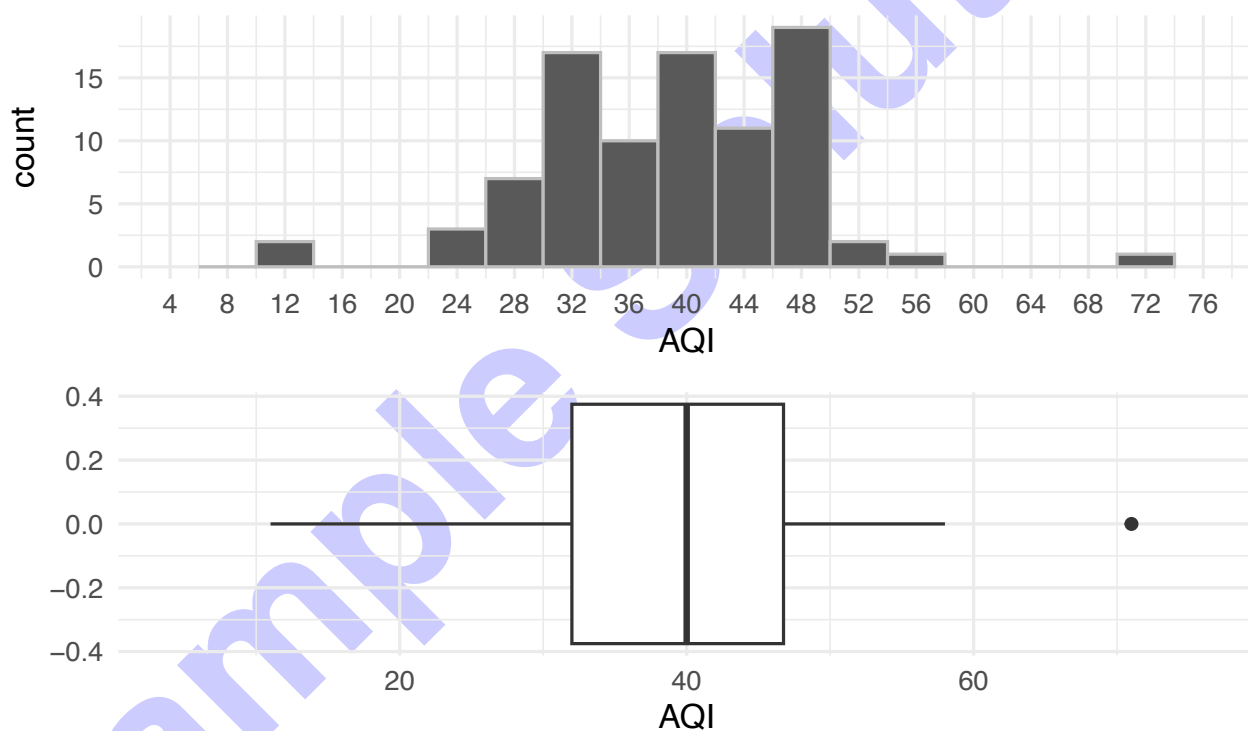
☐ 0
☐
☐ 1
☐ 2

- 0 ☐ b) Decide if the events I (twins are identical) and F (both twins are female) are independent. Justify your answer.

1 ☐

Problem 3 (2 credits)

Daily air quality is measured by the air quality index (AQI) reported by the Environmental Protection Agency. Values of this index range from 0 to 300 and a higher value indicates lower air quality. AQI was reported for 90 randomly sampled days in a town in North Carolina over the past five years. The graphics below show the distribution of these values.



- 0 ☐ a) By eyeballing the given plots, compute the value of the interquartile range.

1 ☐

b) Would you expect the mean of this dataset to be smaller, larger, or roughly equal to the median? Explain your reasoning.

0
1

Problem 4 (8 credits)

The American National Election Studies (ANES) collects data on voter attitudes and intentions as well as demographic information. In this question we will focus on two variables from the 2012 ANES dataset:

- region (levels: Northeast, North Central, South, and West),
- whether the respondent feels things in the country are generally going in the right direction or things have pretty seriously gotten off on the wrong track.

To keep calculations simple we will work with a random sample of 400 respondents from the ANES dataset. The distribution of responses is as follows:

```
addmargins(table(anes))
##               region
## direction  north_central northeast south west Sum
## right_direction      38       21    51   27 137
## wrong_track         60       44   103   56 263
## Sum                98       65   154   83 400
```

According to the 2010 Census, 18% of US residents live in the Northeast, 22% live in the North Central region, 37% live in the South, and 23% live in the West.

a) The goal is to test the hypotheses

H_0 : The sample distribution of regions follows the census distribution.

H_A : The sample distribution of regions does not follow the census distribution

0
1
2

using a goodness of fit test.

Complete the R code below so that this command can be used to simulate the null distribution corresponding to the above hypotheses using the `infer` workflow.

```
P <- c(_____, _____, _____, _____)
names(P) <- c("north_central", "northeast", "south", "west")

null_dist <- anes |>
  specify(response = _____) |>
  hypothesize(
    null = "_____",
    p = P ) |>
  generate(
    reps = 1000,
    type = "draw"
  ) |>
  calculate(stat = "_____")
```

- 0 ☐
- 1 ☐
- 2 ☐
- 3 ☐

b) The observed value of the test statistic for the test in a) is 2.77 (rounded to two digits).

Use the output, given on the right, to test the null hypothesis at the 0.05 significance level and state the test's conclusion in the data context. Justify your answer.

```

null_dist |>
  filter( abs(stat) >= 2.77 ) |>
  summarize( n = n() )

## # A tibble: 1 x 1
##       n
##   <int>
## 1   422

```

- 0 ☐
- 1 ☐

c) In the next step the following statistical analysis was done in R

```
chisq.test(anes$direction, anes$region)
```

```

##
## Pearson's Chi-squared test
##
## data:  anes$direction and anes$region
## X-squared = NA, df = 3, p-value = NA

```

State the null and alternative hypothesis corresponding to the test done in the statistical analysis of the data.
Remark: Be aware that the output is incomplete.

- 0 ☐
- 1 ☐
- 2 ☐

d) Under the null hypothesis of independence between the variables region and direction, the expected counts are equal to

```

##           region
## direction  north_central northeast  south  west
## right_direction    33.565   22.2625  52.745 28.4275
## wrong_track       64.435   42.7375 101.255 54.5725

```

Given these values, do you think the null hypothesis can be rejected or not? Justify your answer.

Problem 5 (5 credits)

Give a short answer to each of the following **unrelated** sub-problems.

a) In a clinical trial, the outcome of a particular medication was compared with that of a placebo: 16 of the 20 patients who took the medicine improved their condition. In contrast, 45 of the 78 in the control group improved. What is the margin of error to estimate the difference in the proportion of subjects that experienced improvement with a 95% confidence interval?

0
1
2

Hint:

```
qnorm(c(0.95, 0.975, 0.99))  
## [1] 1.64 1.96 2.33
```

b) Consider a hypothesis test about two proportions: $H_0 : p_1 = p_2$ versus $H_A : p_1 \neq p_2$. Complete the following sentence by entering in each case one of the following phrases: “a larger, a smaller, an unchanged”, such that the statement is correct.

0
1

Decreasing the sample size in both groups would lead to _____ absolute value of the test statistic and _____ p-value.

c) Ebay sellers wonder if the type of photo posted with an item affects the selling price of that item. One hundred and forty three MarioKart packages were analyzed, which were classified as having a “stock” photo or not. A 95% confidence interval for the average difference in selling price between those without and with “stock” photos ($\mu_{\text{no}} - \mu_{\text{yes}}$) is (-\$7.20, -\$1.14). Decide if you can reject the null hypothesis $\mu_{\text{no}} = \mu_{\text{yes}}$ in favour of the alternative hypothesis $\mu_{\text{no}} \neq \mu_{\text{yes}}$ at a 0.05 significance level. Justify your answer by explaining your reasoning.

0
1
2

Problem 6 (7 credits)

An instructor decided to run three slight variations of the same exam. Prior to handing out the exams, she shuffled the stack of exams to ensure each student received a random version. Anticipating complaints from students, she analyzes the exam scores in the three groups in order to assess whether there is convincing evidence that the level of difficulty was different for at least one of the three versions of the exam. The following are some summary statistics and the ANOVA table associated with this analysis:

```
exam_results |>
  group_by(version) |>
  summarize(q1 = quantile(points, 0.25), q2 = median(points),
            mean = mean(points), q3 = quantile(points, 0.75),
            sd = sd(points), n = n())

## # A tibble: 3 x 7
##   version    q1    q2 mean    q3    sd    n
##   <fct>    <dbl> <dbl> <dbl> <dbl> <dbl> <int>
## 1 a        65.2  73.5  75.1  84.5  13.9   58
## 2 b         63   72   72.0  81.5  13.8   55
## 3 c        71.5  81   78.9  89.5  13.1   51

summary(aov(points ~ version, data = exam_results))
              Df Sum Sq Mean Sq F value Pr(>F)
version         2   1290    645.1   3.484   NA
Residuals     161  29810    185.2

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

0 ☐
1 ☐

a) Compute the sum of squares total SST for the given ANOVA table.

0 ☐
1 ☐
2 ☐
3 ☐

b) The null hypothesis of equal means can be rejected at a significance level of 0.05. Provide an argument for why this is the case.

Hint:

```
qf(c(0.05, 0.95), df1 = 2, df2 = 161)
## [1] 0.05130964 3.05217190
```


c) Since the mean values are not all the same, we want to run pairwise comparisons to analyse which means do differ. The non-adjusted p-values are given in the following table.

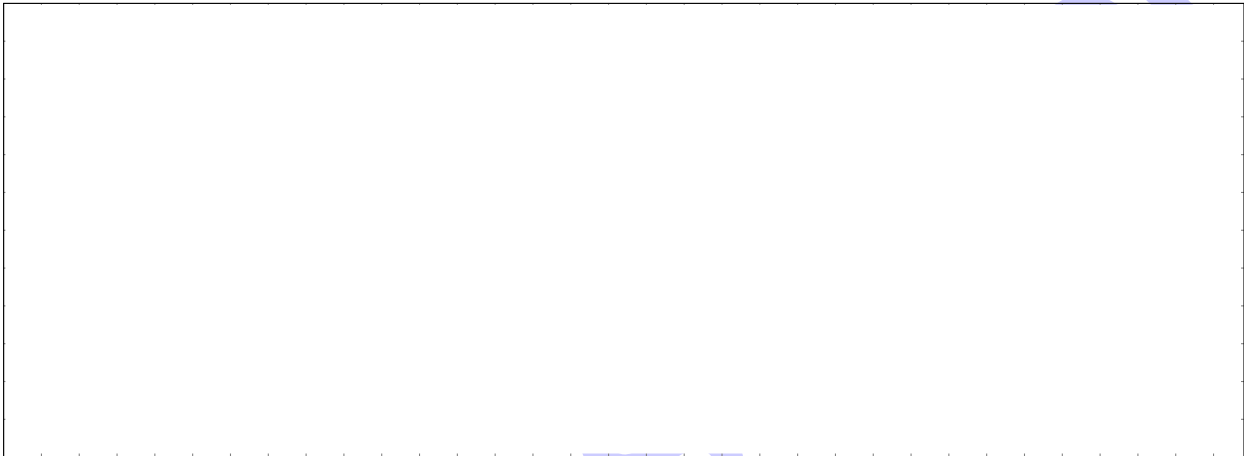
Pairwise comparisons using t tests

data: exam_results\$points and exam_results\$version

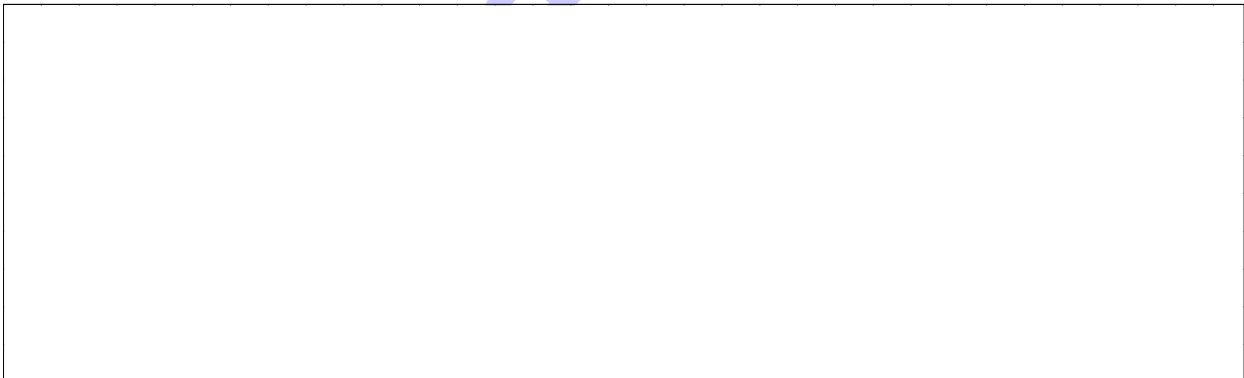
	a	b
b	0.2220	-
c	0.1437	0.0092

P value adjustment method: none

Compute the adjusted p-values based on the Bonferroni method and report all differences, which are significant at a 0.05 level.



d) Suppose the mean score for version b was actually **74.9**, while SST remains unchanged. How, if at all, would this change the F score. Justify your answer.



Problem 7 (7 credits)

The dataset `fat` contains 251 measurements of 14 variables, which can be used to predict the percentage of body fat (`body.fat`). Measuring body fat percentage is relatively expensive compared to measuring the other variables. One would have found a cost-effective alternative to measuring the body fat percentage if a good model fit can be achieved. A model using all predictor variables is fitted with the following command.

```
model_full <- lm(body.fat ~ ., data = fat)
```

- 0 ☐ a) The full model contains uninformative predictor variables, which we want to identify by running a backward selection. In one of the steps, we obtain the following values

1 ☐

```
Step: AIC=696.23
body.fat ~ age + weight + BMI + neck + chest + abdomen + hip +
      thigh + forearm + wrist
```

	Df	Sum of Sq	RSS	AIC
- age	1	85.80	3769.3	700.01
- chest	1	24.92	3708.4	695.92
<none>			3683.5	696.23
- hip	1	27.73	3711.2	696.11
- thigh	1	37.58	3721.1	696.77
- wrist	1	172.90	3856.4	705.74
- forearm	1	40.46	3724.0	696.97
- weight	1	16.01	3699.5	695.32
- neck	1	47.42	3730.9	697.44
- BMI	1	50.68	3734.2	697.66
- abdomen	1	1412.08	5095.6	775.68

Which variable should be removed (if any) in this step? Justify your answer.

Remark: Be aware that the output is not given in the standard form.

- 0 ☐ b) `model_step` is the model obtained by backward selection. For this model the following table contains some part of the summary

1 ☐

2 ☐

3 ☐

```
## # A tibble: 5 x 5
##   term      estimate std.error statistic p.value
##   <chr>      <dbl>      <dbl>      <dbl>    <dbl>
## 1 (Intercept)  3.00         6.86        0.438     NA
## 2 age          0.0748       0.0273      2.74     NA
## 3 BMI          0.513        0.232      2.21     NA
## 4 neck        -0.426       0.205      NA       NA
## 5 thigh        0.170       0.122      NA       NA
```

Decide if age is significant at the significance level 0.05. Justify your answer by approximating the p-value (the values given in the hint allow you to do that). Give an interpretation of the slope estimate for age.

Compute the test statistic for testing if thigh has a significant effect of `body.fat`.

Hint:

```
pt(c(1.651, 1.969), df = 241, lower.tail = FALSE)
## [1] 0.050 0.025
```



c) The adjusted R^2 value of `model_full` is equal to 0.7390968. Decide if the adjusted R^2 of `model_step` is smaller or larger than that. Justify your answer.

Hint:

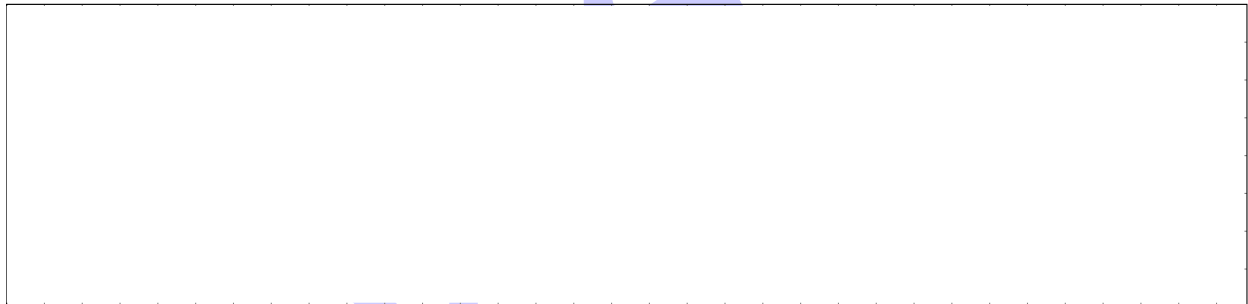
```
AIC( model_full )
```

```
AIC( model_step )
```

```
## [1] 1416.876
```

```
## [1] 1409.622
```

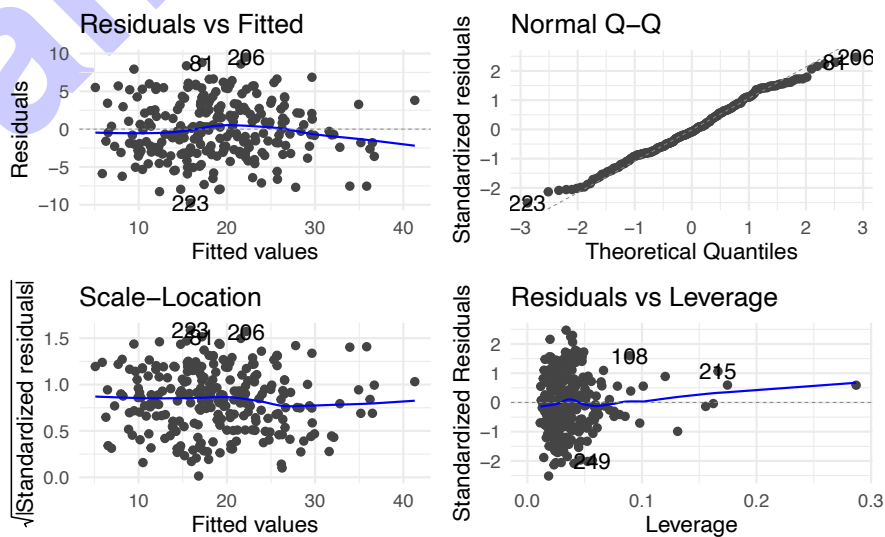
0
1



d) Conduct a residual analysis to check the assumptions of a linear model.

```
autoplot(model_step)
```

0
1
2



Problem 8 (7 credits)

The RMS Titanic sank in the early morning hours of 15 April 1912 in the North Atlantic Ocean. The dataset `titanic` contains information about passengers aboard of the RMS Titanic

`titanic`

A tibble: 714 x 5

Survived Pclass Sex Age SibSp

<int> <int> <chr> <dbl> <int>

1 0 3 male 22 1

2 1 1 female 38 1

3 1 3 female 26 0

4 1 1 female 35 1

5 0 3 male 35 0

6 0 1 male 54 0

7 0 3 male 2 3

8 1 3 female 27 0

9 1 2 female 14 1

10 1 3 female 4 1

i 704 more rows

We are interested in predicting the survival of the passenger (Survived - 1="yes", 0="no") using the variables `Pclass` (Passenger Class - 1, 2, 3), `Sex`, `Age` and `SibSp` (number of sibling/spouses aboard) as predictor variables.

We split the data into a training set `titanic_train` and a test set `titanic_test`. Since we have binary response variable, a logistic regression model will be fitted to the training data.

a) Complete the code given below to fit a logistic regression model using as predictor variables `Pclass`, `Sex`, `Age` `SibSp` and the interaction between `Pclass` and `Sex`.

```
model_train <- _____ (
  Survived ~ _____,
  data = titanic_train,
  family = _____)
```

b) A summary of the fitted model is the following

A tibble: 6 x 5

##	term	estimate	std.error	statistic	p.value
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	(Intercept)	8.50	1.10	7.75	9.43e-15
## 2	Pclass	-2.43	0.364	-6.67	2.49e-11
## 3	Sexmale	-6.22	1.02	-6.08	1.23e-9
## 4	Age	-0.0466	0.00934	-5.00	5.88e-7
## 5	SibSp	-0.355	0.131	-2.72	6.50e-3
## 6	Pclass:Sexmale	1.52	0.385	3.96	7.64e-5

Write out the form of the fitted model for a **male passenger**. Simplify the expression as much as possible.

☐ 0
☐ 1
☐ 2

c) Give an **interpretation** of the estimated slope of Pclass in terms of the odds ratio for a **male passenger**.
Remark: Computing the odds ratio is necessary but insufficient. You should describe in words the meaning of the computed value.

☐ 0
☐ 1

d) Given model_train, we are now able to compute predictions for the test dataset.

```
titanic_test$pred_prob <- predict(model_train, newdata = titanic_test, type = "response")
```

☐ 0
☐ 1

To predict survival we need a threshold for the computed probabilities of survival. Choosing a threshold of 0.5 leads to a sensitivity of 0.72. Compute the number of true positives.

Hint:

```
table(titanic_test$Survived)
```

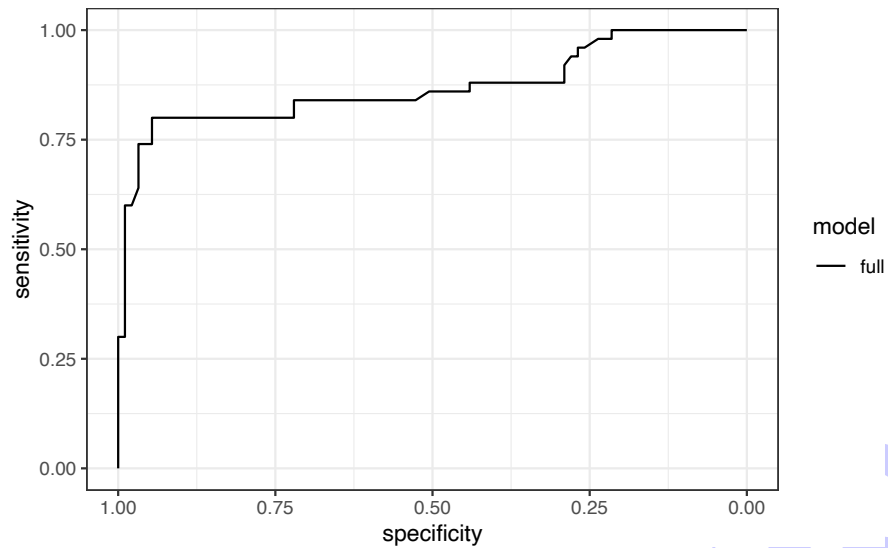
```
##
```

```
## 0 1
```

```
## 93 50
```

0 ☐
1 ☐

e) The following graphic contains the ROC curve for the predictions `titanic_test$pred_prob`.



Explain what is measured by the AUC value (in this case, equal to 0.8691398).

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Cheat sheet

Exploring data:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad s = \sqrt{s^2} \quad IQR = Q_3 - Q_1 \quad Q_1 - 1.5 \cdot IQR, \quad Q_3 + 1.5 \cdot IQR$$

$$r_{(x,y),n} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{s_x \cdot s_y}$$

Probability:

$$P(A^c) = 1 - P(A); \quad P(A \cap B) = P(A) \cdot P(B), \quad A, B \text{ independent}; \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}; \quad E(X) = \sum_{i=1}^k x_i P(X = x_i), \quad E(X) = \int_{\mathbb{R}} x \cdot f(x) dx; \quad \text{Var}(X) = E(X^2) - E(X)^2;$$

$$SD(X) = \sqrt{\text{Var}(X)}; \quad \text{Cov}(X, Y) = E(X \cdot Y) - E(X) \cdot E(Y); \quad Z = \frac{X - E(X)}{\sqrt{\text{Var}(X)}}$$

$$\text{Geom. dist.: } P(X = x) = (1 - p)^{x-1} p, \quad x \in \mathbb{N}; \quad E(X) = \frac{1}{p}, \quad SD(X) = \sqrt{\frac{1-p}{p^2}}$$

$$\text{Binom. dist.: } P(X = x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}, \quad x \in \{0, 1, 2, \dots, n\}; \quad E(X) = np, \quad SD(X) = \sqrt{np(1-p)}$$

$$\text{Neg. Binom. dist.: } P(X = x) = \frac{(x-1)!}{(k-1)!(x-k)!} p^k (1-p)^{x-k}, \quad x \geq k$$

$$\text{Pois. dist.: } P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x \in \mathbb{N}_0; \quad E(X) = \lambda, \quad SD(X) = \sqrt{\lambda}$$

Inference for categorical data:

$$\hat{p} \pm z^* \cdot SE(\hat{p}) \quad SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \cdot SE(\hat{p}_1 - \hat{p}_2) \quad SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad \hat{p}_{\text{pool}} = \frac{\hat{p}_1 n_1 + \hat{p}_2 n_2}{n_1 + n_2}$$

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}, \quad \text{expected}_{ij} = \frac{\text{row } i \text{ total} \cdot \text{column } j \text{ total}}{\text{table total}} \text{ (ind.)}, \quad df = k - 1 \text{ or } (r - 1) \cdot (c - 1)$$

Inference for numerical data:

$$\bar{x} \pm t_{df}^* \cdot SE(\bar{x}) \quad T_{df}(\mathbf{x}) = \frac{\bar{x} - \mu_0}{SE(\bar{x})} \quad SE(\bar{x}) = \frac{s(x)}{\sqrt{n}} \quad df = n - 1$$

$$(\bar{x} - \bar{y}) \pm t_{df}^* \cdot SE(\bar{x} - \bar{y}) \quad T_{df}(\mathbf{x}, \mathbf{y}) = \frac{(\bar{x} - \bar{y}) - 0}{SE(\bar{x} - \bar{y})} \quad SE(\bar{x} - \bar{y}) = \sqrt{\frac{s(x)^2}{n_1} + \frac{s(y)^2}{n_2}} \quad df = \min(n_1 - 1, n_2 - 1)$$

$$\text{two-samples: } n \geq \frac{2 \cdot \sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2; \quad \text{one-sample: } n \geq \frac{\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2 \text{ with } \alpha^* \in \{\alpha, \alpha/2\}$$

ANOVA and linear regression:

$$F = \frac{MSG}{MSE} \quad df_1 = k - 1, \quad df_2 = n - k, \text{ for } k \text{ groups}; \quad SSG(\mathbf{y}) = \sum_{i=1}^k n_i (\bar{y}_i - \bar{y})^2, \quad SST(\mathbf{y}) = \sum_{j=1}^n (y_j - \bar{y})^2,$$

$$SSE(\mathbf{y}) = SST(\mathbf{y}) - SSG(\mathbf{y}); \quad p_{adj,i} = K \cdot p_i$$

$$\hat{y} = b_0 + b_1 x \quad b_1 = r_{(x,y),n} \cdot \frac{s_y}{s_x} \quad \bar{y} = b_0 + b_1 \bar{x} \quad \text{residual} = y - \hat{y} \quad \hat{\sigma} = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2}$$

$$T_{df}(\mathbf{y}, \mathbf{x}) = \frac{b_1(\mathbf{y}, \mathbf{x}) - \beta_{1,0}}{SE_{b_1}(\mathbf{y}, \mathbf{x})} \quad b_1(\mathbf{y}, \mathbf{x}) \pm t_{df}^* \cdot SE_{b_1}(\mathbf{y}, \mathbf{x}) \quad df = n - 2; \quad R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$R_{adj}^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2 / (n - k - 1)}{\sum_{i=1}^n (y_i - \bar{y})^2 / (n - 1)}; \quad AIC = \text{constant} + n \cdot \ln \left(\sum_{i=1}^n e_i^2 \right) + 2k$$

Logistic regression:

$$\text{logit}(p_i) = \log \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i} \quad OR_j = \frac{\text{odds when } x_{i,j}=x+1}{\text{odds when } x_{i,j}=x} = e^{\beta_j}$$

$$\text{sensitivity} = P(\text{Test} = + | \text{Condition} +), \quad \text{specificity} = P(\text{Test} = - | \text{Condition} -)$$

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

A large grid of graph paper, consisting of 30 columns and 30 rows of small squares. A diagonal watermark with the text "Sample Solution" in a light blue, sans-serif font is overlaid across the grid, running from the bottom-left towards the top-right.